

B. Wieneke

Stereo-PIV using self-calibration on particle images

Received: 25 October 2004 / Revised: 14 February 2005 / Accepted: 1 March 2005 / Published online: 26 May 2005
© Springer-Verlag 2005

Abstract A stereo-PIV (stereo particle image velocimetry) calibration procedure has been developed based on fitting a camera pinhole model to the two cameras using single or multiple views of a 3D calibration plate. A disparity vector map is computed on the real particle images by cross-correlation of the images from cameras 1 and 2 to determine if the calibration plate coincides with the light sheet. From the disparity vectors, the true position of the light sheet in space is fitted and the mapping functions are corrected accordingly. It is shown that it is possible to derive accurate mapping functions, even if the calibration plate is quite far away from the light sheet, making the calibration procedure much easier. A modified 3-media camera pinhole model has been implemented to account for index-of-refraction changes along the optical path. It is then possible to calibrate outside closed flow cells and self-calibrate onto the recordings. This method allows stereo-PIV measurements to be taken inside closed measurement volumes, which was not previously possible. From the computed correlation maps, the position and thickness of the two laser light sheets can be derived to determine the thickness, degree of overlap and the flatness of the two sheets.

1 Introduction

For stereo-PIV (stereo particle image velocimetry), correct calibration is an essential prerequisite for measuring accurately the three velocity components. Most often, an empirical approach is used by placing a planar calibration target with a regularly spaced grid of marks at exactly the position of the light sheet and moving the target by a specified amount in the out-of-plane

direction to two or more z positions (Soloff et al. 1997). At each z position (light sheet plane defined by $z=0$), a calibration function with sufficient degrees of freedom maps the world x - y plane to the camera planes, while the difference between the z planes provides the z derivatives of the mapping function necessary for reconstructing the three velocity components. This empirical approach has the advantage that all image distortions arising from imperfect lenses or light path irregularities (e.g. from air/glass/water interfaces) are compensated for automatically in one step.

Alternatively, one can use a 3D calibration plate with marks on two z levels, avoiding the need for rigid mechanical setups with accurate translation stages. Different mapping functions have been used, from a second-order or third-order polynomial in x and y (Soloff et al. 1997) to functions derived from the perspective equations (camera pinhole model) (Willert 1997). A major drawback of this empirical method is the need to position the calibration plate exactly at the same position as the light sheet, which is often very difficult to accomplish. A correction scheme based on a cross-correlation between the image of cameras 1 and 2 has been proposed (Willert 1997; Coudert and Schon 2001), which is also the basis for the calibration correction method proposed in this work.

In parallel, especially with the advent of inexpensive digital cameras, extensive work has been done in the field of computer vision and photogrammetry to compute accurate camera calibrations. While only a 2D mapping function with additional z derivatives is required for stereo-PIV with thin light sheets, computer vision, in general, requires a volume mapping function to map all x - y - z -world-points to the recorded x - y -pixel-locations on one or more cameras. Usually, this is done with a camera pinhole model with added parameters for lens distortions (Tsai 1986).

There are six external projective parameters mapping the calibration plate by a rotation and translation to the world camera plane perpendicular to the optical axis and internal camera parameters, like the focal length, the

B. Wieneke
LaVision GmbH, Anna-Vandenhoeck-Ring 19,
37081 Göttingen, Germany
E-mail: b.wieneke@lavision.de

principal point, which is the foot point of the optical axis onto the CCD, the pixel size and radial lens distortion terms. The optical axis is defined as the line perpendicular to the CCD chip passing through the pinhole.

A variety of 2D calibration and 3D calibration targets have been used successfully. A common calibration method consists of recording a known planar calibration target at a few (4–8) shifted and rotated positions. This is done either by moving a single camera or a stereo rig setup with two cameras around a fixed target (“walk-around problem”) or by having the cameras fixed and moving the target. All fixed parameters (internal camera parameters and the relative position and orientation of the two cameras), together with all parameters unique for the particular view, are fitted by a nonlinear least-squares fit (bundle adjustment). Due to a very sparsely populated Hessian matrix, since the external parameters unique for each view are independent of each other, special provisions are incorporated in the fit algorithm for better numerical convergence, less computing time and higher accuracy. A good overview of self-calibration methods and bundle adjustment fits is given by Hartley and Zissermann (2000).

In the present work, a stereo-PIV calibration method has been implemented and tested using the camera pinhole model. Instead of requiring a perfect alignment between the calibration plate and the light sheet, a correction scheme has been developed that provides accurate mapping functions, even when the calibration plate is quite far away or tilted relative to the light sheet. This makes the stereo-PIV calibration easier and more accurate. Similar work has been done before on a setup with telecentric lenses (Fournel et al. 2003) and with standard lenses with a Scheimpflug adapter (Fournel et al. 2004).

An important application of self-calibration is the case where it is difficult or even impossible to place a calibration plate inside a closed measurement volume. In these cases, one would ideally like to calibrate the cameras outside the flow apparatus and self-calibrate onto the light sheet inside. For these cases, different strategies are presented in Sect. 4, together with experimental validation.

2 Self-calibration method

2.1 Camera pinhole model

For stereo-PIV, two mapping functions need to be determined: $\mathbf{M}_1$ for camera 1 and $\mathbf{M}_2$ for camera 2, relating a world coordinate $\mathbf{X}_w = (X_w, Y_w, Z_w)$ to pixel locations $\mathbf{x}_1 = (x_1, y_1)$ and $\mathbf{x}_2 = (x_2, y_2)$ in the recorded images of cameras 1 and 2 (Fig. 1):

$$\mathbf{x}_1 = \mathbf{M}_1(\mathbf{X}_w) \text{ and } \mathbf{x}_2 = \mathbf{M}_2(\mathbf{X}_w) \quad (1)$$

In the empirical approach with a calibration plate at two or more z positions (Soloff et al. 1997), it is sufficient to

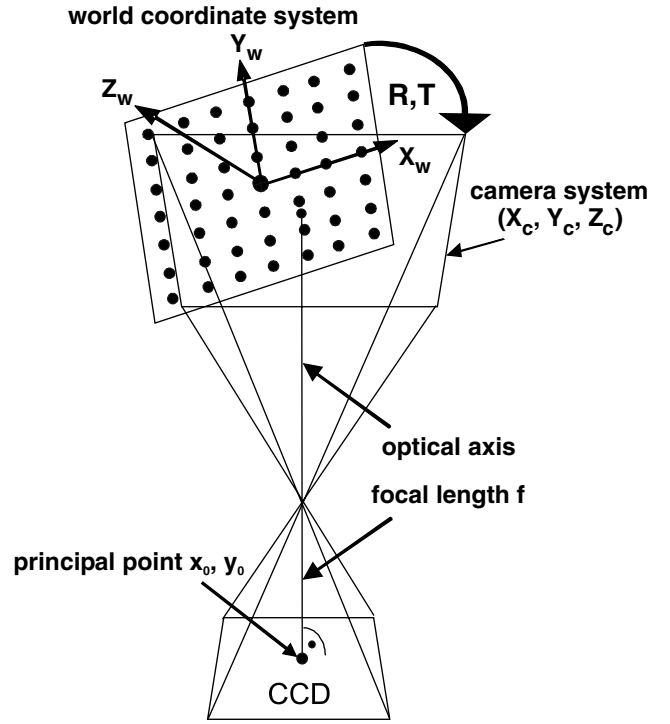


Fig. 1 Camera pinhole model

know $\mathbf{M}(X_w, Y_w, Z_w=0)$ and the z derivatives $\delta x_i / \delta Z_w$ and $\delta y_i / \delta Z_w$, $i=1, 2$ of the mapping function at $(X_w, Y_w, Z_w=0)$, which do not change significantly across the thickness of the light sheet.

In contrast, the camera pinhole model provides a complete mapping of the volume as given by Eq. 1. The camera pinhole model used is based on Tsai’s 11-parameter model (Tsai 1986). The six external camera parameters are given by the rotation $\mathbf{R}$ and translation $\mathbf{T}$ of the world coordinates $\mathbf{X}_w$ to the camera coordinates $\mathbf{X}_c = (X_c, Y_c, Z_c)$:

$$\mathbf{X}_c = \mathbf{R} \mathbf{X}_w + \mathbf{T} \quad (2)$$

The undistorted and distorted camera position $\mathbf{x}_u = (x_u, y_u)$ and $\mathbf{x}_d = (x_d, y_d)$, respectively, are computed by:

$$\begin{aligned} x_u &= f \frac{X_c}{Z_c} \\ y_u &= f \frac{Y_c}{Z_c} \end{aligned} \quad (3)$$

with:

$$\begin{aligned} x_d &= x_u (1 + k_1 r + k_2 r^2) \\ y_d &= y_u (1 + k_1 r + k_2 r^2) \end{aligned} \quad (4)$$

and:

$$r^2 = x_d^2 + y_d^2$$

where k_1 and k_2 are the first-order and second-order radial distortion terms, respectively. Usually, for good quality lenses, k_1 is sufficient; for wide-angle lenses or consumer cameras, additional radial terms and even

tangential distortion terms may be required. f is the distance between the principal point and the camera pinhole. This is close to but not exactly the focal length of the lens f_{lens} . Provided that the calibration plate is in focus at distance T_z , f is theoretically related to f_{lens} by:

$$f = f_{\text{lens}} \left(1 + \frac{f_{\text{lens}}}{T_z + f_{\text{lens}}} \right) \quad (5)$$

but in practice, lenses specified with a fixed focal length do not exactly obey this equation. For good lenses, the deviations are on the order of 2–5%.

Finally, the true pixel location $\mathbf{x} = (x, y)$ on the CCD chip is given by:

$$\begin{aligned} x &= S_x \frac{x_d}{S_{\text{pixel}}} + x_0 \\ y &= \frac{y_d}{S_{\text{pixel}}} + y_0 \end{aligned} \quad (6)$$

S_x is an optional skew factor, which is 1 for a square pixel. Otherwise, S_x can be used as an additional distortion term. S_{pixel} is the (known) pixel size. Only the ratio of f/S_{pixel} is relevant. (x_0, y_0) is the principal point, which is usually close to the middle of the CCD chip, unless a Scheimpflug adapter is used. The set of parameters (f, S_x, k_1, x_0, y_0) constitute the five internal camera parameters of the Tsai's model.

While a possible Scheimpflug angle is included in the pinhole model, it might lead to radial distortion parameters, which are no longer rotationally symmetric, since the optical axis of the pinhole model no longer coincides with the optical axis of the camera lens. Fournel et al. (2004), therefore, proposed a special Scheimpflug rotation in the pinhole model.

The world coordinates $\mathbf{X}_w$ are given in some physical units (e.g. millimetres). For dewarping images and the actual stereo-PIV computation, it is necessary to convert the world coordinates to pixel units. This can be done in an arbitrary way. In the following computations, the conversion scale of millimetres to pixels is set in such a way that, in both directions, the size of a dewarped pixel is equal to or smaller than in the raw image in order not to lose any information by pixel binning. The size of the dewarped image is chosen such that it corresponds to the same part of the field-of-view visible by both cameras. The coordinate origin is initially defined by the position of one of the marks on the calibration plate.

For two cameras, about 22 parameters are needed to specify the pinhole model. This compares to 80 parameters for the third-order 2D polynomial defined on two z planes. With only a few correction terms, the pinhole function has proven to be highly accurate in the field of computer vision. Therefore, it is obvious that not all 80 parameters of the third-order polynomial are statistically significant. In general, it is better to start with an appropriate physical model and then add a few significant correction terms. Otherwise, one might fit noise e.g. in the detected position of the marks on the calibration plate. The third-order polynomial is more appropriate in cases of strong non-perspective global or local

distortions (e.g. viewing through cylinders, where it is more difficult to design appropriate correction terms for the pinhole model). Since the pinhole model is highly nonlinear, it is important that additional terms are fairly orthogonal to the other parameters. Otherwise, the fit would not converge properly.

2.2 Correction of mapping function

Whether or not the alignment of the laser sheet and calibration plate is perfect can be checked by dewarping the camera images with $\mathbf{M}^{-1}$ at $Z_w = 0$ and comparing the two images. Ideally, the two images should match perfectly. Any deviation d (Fig. 2) (“disparity map”) is an indication of some misalignment, leading to different error sources. If the calibration plate has a z offset relative to the measurement plane, the reconstruction of the three-component (3C) vector $(\mathbf{u}, \mathbf{v}, \mathbf{w})$ from the two two-component (2C) vectors $(\mathbf{u}_i, \mathbf{v}_i)$ uses vectors computed at different locations in the measurement plane. For example, with an offset of 20 pixels and a velocity gradient of 5% (in units of pixel displacement per pixel distance), the displacement would change by 1 pixel over a distance of 20 pixels, which leads to considerable errors in the reconstructed 3C vector. Also, slightly incorrect derivatives of the mapping functions would be used. Finally, the coordinate system itself remains warped. That is, a tilt of the calibration plate by 1° means that the calculated 3C vectors are rotated by 1° .

This disparity map has been used by Willert (1997) for correcting the position at which the corresponding 2C vectors are calculated for cameras 1 and 2. A more advanced approach has been created for the case of telecentric lenses recomputing a new mapping function from the disparity map (Fournel et al. 2003) and, more recently, for normal lenses with Scheimpflug adapters (Fournel et al. 2004). A similar general approach is used here in a series of different steps.

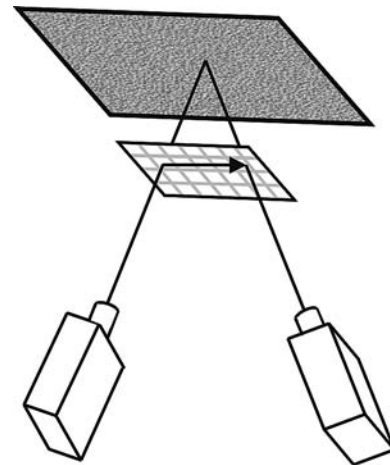


Fig. 2 The world points of the light sheet can be computed by triangulation using the disparity vectors

Firstly, the computation of the disparity vector map of the two dewarped images is computed by a standard cross-correlation PIV technique. The particle pattern inside an interrogation window looks quite different when viewed from the two camera viewpoints, since the particles are dispersed throughout the light sheet. Therefore, it is usually insufficient to simply correlate a single image pair. Instead, an ensemble-averaging algorithm is used by summing the correlation planes of many image pairs (Meinhart et al. 1999). Depending on the particle density and the thickness of the light sheet, about 5–50 images are typically needed to compute an accurate disparity map from a well-shaped correlation peak. For large fields of view (e.g. in wind tunnels), a single view might be sufficient, which offers the potential to correct vibrational displacements of the laser sheet or the cameras (Willert 1997). Multi-pass algorithms with deformed interrogation windows can be applied to further enhance the accuracy of the vector map. This means that all n images are processed first to arrive at an initial guess of the disparity vector map, which is then used as a reference vector field to shift and deform the interrogation windows in the next pass and so on.

Willert (1997) used these disparity vectors to correct the position where the vectors are computed. For small misalignments, this effectively removes the main error source of computing the vectors of cameras 1 and 2 at different positions. In case of larger misalignments between the plate and the light sheet, a more advanced volume correction scheme must be used to compute a correct coordinate system of the light sheet plane with accurate spatial derivatives of the mapping function, as explained below.

Once the disparity map has been computed, a corresponding world point in the measurement plane is computed by a standard triangulation method for each vector (Fig. 2). Errors in the disparity vectors mean that the two reprojected lines from each camera do not exactly intersect at a single world point. It is optimal to find a point in space whose projections onto the two camera images are closest to the measured positions (Hartley and Sturm 1994). These criteria can be used to eliminate false disparity vectors using a sensible threshold (e.g. 0.5 pixels). Instead of working with dewarped images, one can also compute the disparities between the original images and use them in the triangulation step to arrive at the same world points. The advantage of using dewarped images is only that the PIV user can check for remaining disparities (non-zero vectors) more easily. Triangulation is only possible when one has a volume mapping function, but it must not necessarily be a pin-hole mapping function. It can also be an accurate empirical third-order 2D polynomial function calculated for a number of parallel z planes, which cover enough of the volume to incorporate the laser light sheet to be fitted.

A plane is then fitted through the world points in 3D space and the mapping functions of cameras 1 and 2 are corrected by a corresponding transformation such that

the fitted measurement plane becomes the $z=0$ plane. This is done by replacing, in Eq. 2, R by $R \times dR$ and T by $T + (R \times dT)$, where dR is the rotation of the fitted plane relative to the calibration plate and dT is the distance of the plane to the calibration plate. The freedom of choosing the new coordinate system due to in-plane rotation and choice of origin is reduced by setting the new origin as the point projected from the previous origin in camera 1 onto the measurement plane and, also, the previous x axis of camera 1 coincides with the x axis of the measurement plane. This can be changed later by the user to set a new origin and x axis or y axis in a dewarped particle image. The complete correction scheme is shown in Fig. 3.

The whole procedure can be repeated again to arrive at better fits. Usually, the process has converged and the disparity map does not get smaller after two or three passes. Good results have been achieved by using only a single-pass cross-correlation and repeating the complete correction process two or three times.

The triangulation error and the error from fitting a plane through the world points provide information about the quality of the fit. Both errors are affected by inaccuracies in the disparity vectors, which are mostly random errors due to an insufficient number of particles and the well-known bias errors of the correlation function, together with systematic calibration errors in case the computed mapping function becomes inaccurate when projected in space towards the light sheet plane. Calibration errors often lead to high triangulation errors, while the plane fit error might remain small. As shown later, with a good setup, it is possible to compute

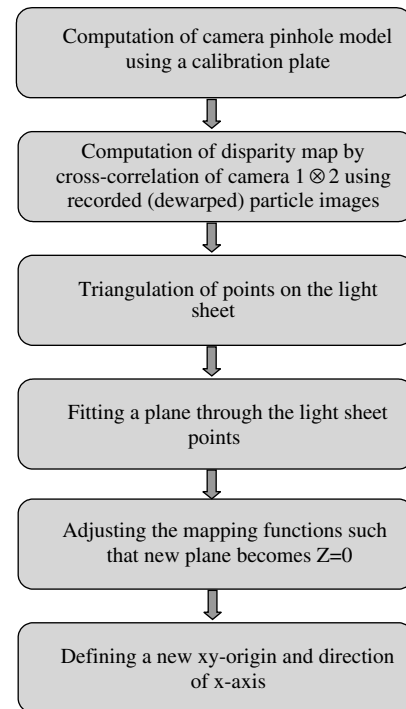


Fig. 3 Flow chart of self-calibration procedure

the position of the light sheet to within 0.1 pixels of the centre of the sheet, with a thickness of typically 10–20 pixels—something that is hardly possible by simple visual inspection and manual placement of the plate.

The correction scheme above assumes that the internal camera parameters, as well as the position and orientation of camera 2 relative to camera 1, do not change. In the 22 parameters of the stereo Tsai model, one can substitute the external parameters $\mathbf{R}_2$ and $\mathbf{T}_2$ of camera 2 by the relative transformation $\mathbf{R}_{12}$ and $\mathbf{T}_{12}$ of camera 2 relative to camera 1. Then, the self-calibration procedure above is equivalent to newly fitting $\mathbf{R}_1$ and $\mathbf{T}_1$, whilst keeping $\mathbf{R}_{12}$ and $\mathbf{T}_{12}$ fixed. Since the coordinate origin and the x axis are freely selectable, this leaves three parameters of $\mathbf{R}_1$ and $\mathbf{T}_1$ to be fitted, which are the three parameters of the position and orientation of a plane in space. Therefore, the procedure of triangulation, plane fit and transformation can be replaced by a single non-linear fit of the three free parameters of $\mathbf{R}_1$ and $\mathbf{T}_1$ using the relationship between the disparity and the mapping function parameter given by the well-known fundamental equation (Hartley and Zissermann 2000):

$$\mathbf{x}_1^T \mathbf{F} \mathbf{x}_2 = 0 \quad (7)$$

where $\mathbf{F}$ is the fundamental 3×3 matrix of rank 2 with 8 degrees of freedom and $\mathbf{x}_1 = (x_1, y_1, 1)$ and $\mathbf{x}_2 = (x_2, y_2, 1)$ are the camera coordinates. It is, nevertheless, quite instructive to perform the three steps separately to identify the different error sources and to check the flatness of the light sheet.

Advanced self-calibration methods can be devised to fit more than the three plane parameters, since the fundamental equation has 8 degrees of freedom. One might fit user-adjusted focal lengths or Scheimpflug positions or even the relative position between cameras 1 and 2 with some restrictions. This is a subject of further research.

2.3 Stereo-PIV processing and 3C reconstruction

Different approaches for stereo vector computation have been proposed, as summarised e.g. by Prasad (2000) or Callaud and David (2004). In all cases, a 2D2C vector field is computed for each camera from which a 2D3C vector field is computed by stereoscopic reconstruction. One has the choice of:

1. Computing the 2D2C field on a regular grid in the raw images and using the two interpolated vectors to compute a 3C vector at regular world grid positions
2. Computing the 2D2C vectors in the raw image at a position corresponding to the correct world position
3. Dewarping the images first and computing the 2D2C vectors at the correct world grid position

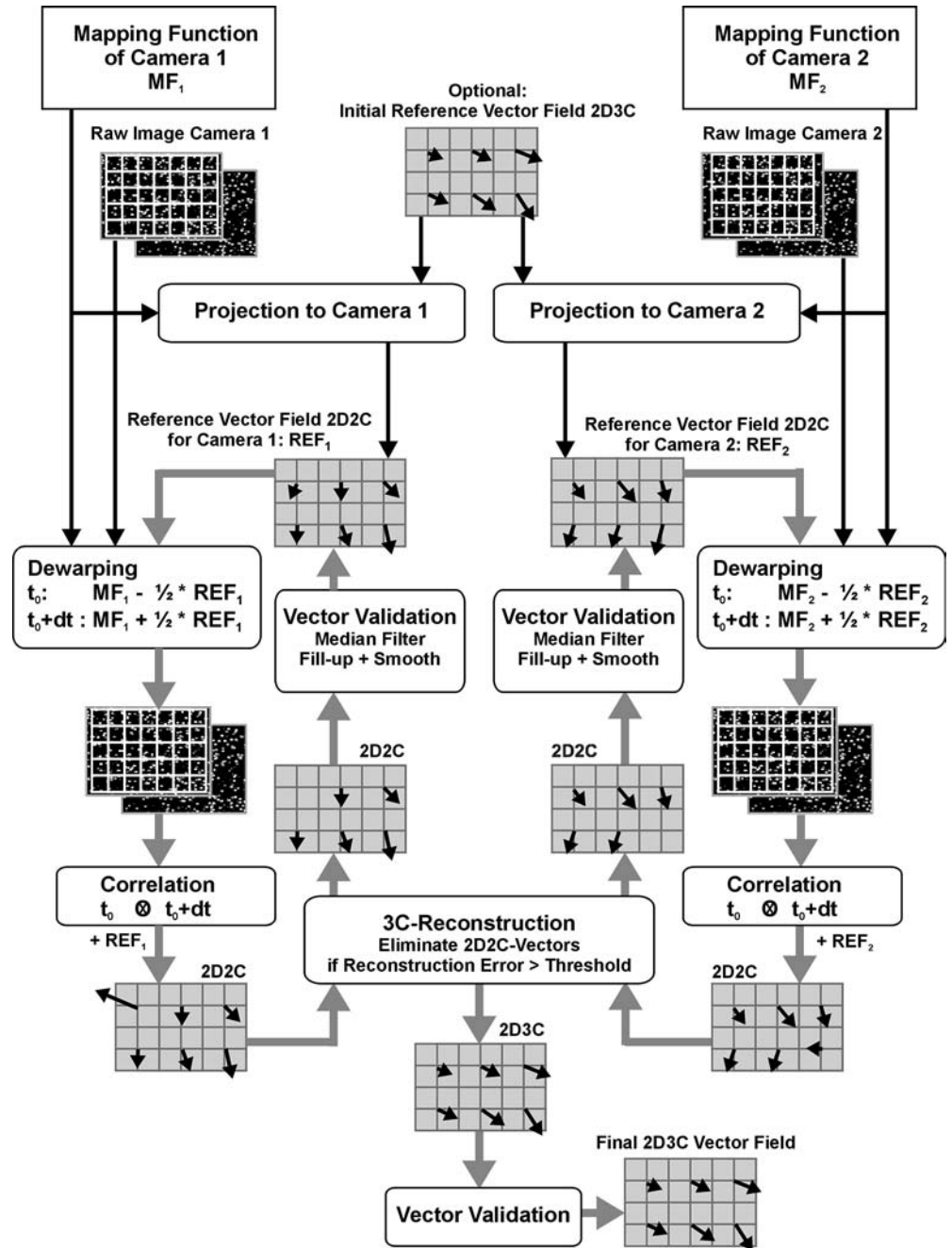
Method 1 has the disadvantage that the vectors are not computed at the correct world position and, due to vector interpolation, false or inaccurate vectors affect four final 3C vectors. Method 2 has the disadvantage

that the size and shape of the interrogation windows differ between the two cameras, due to the perspective viewing. For method 3, the computed 2D2C vectors are already computed at the world correct position and originate from the same interrogation window of equal size and shape, but a sub-pixel interpolation is required during the dewarping, which, together with the sub-pixel interpolation necessary for the multi-pass window deformation scheme, leads to added image degradation. Therefore, a modified method 3 approach is used here, where the dewarping and image deformation is done simultaneously before each step in the multi-pass iterative scheme.

For the first computational pass, the two frames ($t_0, t_0 + dt$) of each camera are dewarped and evaluated. This already provides vectors at the correct position in the world coordinate system. Also, the size and shape of the interrogation windows for both cameras are the same, which means that the correlation is done on the same particles, apart from the effects due to the non-zero thickness of the light sheet. Then, a preliminary 3C reconstruction is done to remove corresponding vectors in the 2C vector fields for which the reconstruction error is too large (e.g. larger than 0.5 or 1 pixel). This method very effectively removes spurious vectors, since two false vectors with random directions are rarely correlated. At the end of the first pass, missing vectors are interpolated and the vector field is smoothed slightly for numerical stability.

The resulting vector field is used as a reference for deforming the interrogation windows in the next pass. Actually, not every interrogation window is deformed individually, but the complete image is deformed at once, with half the displacement in the backwards direction assigned to the first image at t_0 and the other half in the forwards direction to the second image at $t_0 + dt$. Image deformation requires less floating-point operations, since e.g. for an overlap of 75%, the same region would be deformed 16 times using window deformation. Image deformation is combined with the dewarping of the original image in one step. Usually, after three or four passes at the final interrogation window size, the 2D2C vector fields have converged sufficiently. Then, the 3C reconstruction is performed, which consists of solving a system of four linear equations with three unknowns (u, v, w). This is done by using the normal equation, which distributes the error evenly over all three components. Computing from (u, v, w) the (u_1, v_1) and (u_2, v_2) components, the deviation from the measured (u_1, v_1) and (u_2, v_2) can be calculated (“reconstruction error”). Usually, with a good calibration and 2C vector errors of less than 0.1 pixels, the reconstruction error is well below 0.5 pixels. This can be used as an efficient rejection of false random vectors, which usually produce large reconstruction errors. The complete flow chart is shown in Fig. 4. The 2D2C vector fields are separately computed for cameras 1 and 2. A typical multi-pass scheme consists, for example, of one or two passes with an interrogation window size of 64×64 pixels and 50% overlap, followed by four passes

Fig. 4 Flow chart of stereo-PIV vector field computation



with a window size of 32×32 and 75% overlap. After each pass, a 2D3C reconstruction is done for the purpose of eliminating the 2D2C vectors with reconstruction errors above some predetermined threshold (e.g. 1 pixel), but the 2D3C vector field is not used further. Only the data at the end the reconstructed 2D3C field is taken and validated e.g. by a median filter.

3 Experiments

In Sect. 3.1, 16 different calibrations have been taken and self-calibrated on a recording of a flat random

pattern plate. A bundle adjustment of the closest eight calibrations serves as a reference. The corrected calibrations are compared in different ways to assess the different residual errors after self-calibration. In Sect. 3.2, a flat random pattern plate has been moved by a translation stage and the measured stereo-PIV displacements are compared to the true displacement. This is for calibration and self-calibration done in air, as well as in water, to verify the accuracy of the pinhole model when used with intermediate index-of-refraction changes. In Sect. 3.3, a real experiment in air has been performed and the vector fields before and after correction are computed and analysed.

3.1 Experimental results with synthetic images

Sixteen views of a 3D calibration plate with a size of about 100×100 mm are recorded at different positions and orientations. The image size is 1,280×1,024 pixels, and a small aperture with an f stop of 20 has been used to ensure a large depth of focus. From each view, a camera pinhole mapping function is calculated with fixed $s_x = 1$ and $k_2 = 0$. Table 1 shows some of the pinhole parameters for camera 1.

A flat plate with a random dot pattern is recorded, defining a light sheet position. This image is used later for self-calibration. A reference mapping function is computed by a bundle adjustment of the eight calibration views closest to the random pattern plate. Note that the parameters for calibration 1 are almost the same as for the reference, since the first of the eight views for the bundle adjustment is taken as the reference coordinate system $z = 0$.

Table 1 shows the average length of the disparity vectors computed by cross-correlation between the random dot image of camera 1 and camera 2 for all calibrations. Even extreme disparities of up to 500 pixels are present, which means that only half of the calibration plate was visible to both cameras simultaneously.

The self-calibration procedure of Sect. 2.2 has been applied to the reference mapping function. In the following, the other mapping functions are compared to the corrected reference function in different ways.

A synthetic double-frame random pattern image has been generated based on a 2D2C velocity field of regular vortices with an average gradient of about 10% and ± 5 -pixel displacement. Seeding density is high, particle diameter is around 2 pixels and no noise is included. The two frames are warped to $z = 0$ using the corrected reference mapping function, leading to a 4-frame-reference stereo-PIV image, two frames warped with camera 1

mapping function, two with camera 2 function. Since all particles are located at $z = 0$, errors due to particles spaced throughout a light sheet are avoided. Evaluating this image with the stereo-PIV procedure using the corrected reference mapping function gives back the original 2D3C vector field with a zero out-of-plane w component.

The still uncorrected 16 mapping functions are used to compute a stereo-PIV 2D3C vector field from the synthetic 4-frame image in the usual way, including final vector validation. The final interrogation window size is 32×32 pixels, with an overlap of 75%. For a disparity of only a few pixels, the errors are still quite small and only few false vectors occur. Calibration 3 with 9-pixel disparity shows an error of 0.44 pixels, as is roughly expected from disparity×velocity gradient. For all larger disparities, a meaningful velocity field can no longer be calculated. All vectors are eliminated either by the reconstruction error filter, which throws out all vectors with a reconstruction error above 1 pixel, or by the final vector validation using a regional median filter.

In the next step, the mapping functions 1–16 are corrected using the self-calibration procedure on the recorded random pattern image. For the corrected mapping functions, the average residual disparity is computed again (Table 2). It is usually well below 0.1 pixels, even for very high initial misalignments. Most of the remaining errors are due to standard correlation errors. Only calibrations 10, 12, 14 and 15 have a higher residual misalignment relative to the reference calibration.

The corrected mapping functions are compared to the corrected reference mapping function, firstly with a direct comparison of the functional form of the mapping functions. Using a grid of 10×10 image points in the $z = 0$ plane, the values of the mapping functions are computed and compared to the values of the corrected reference

Table 1 Mapping function parameters before correction

Calibration	f_1 (mm)	T_{z1} (mm)	R_{x1} (°)	R_{y1} (°)	R_{z1} (°)	Disparity (pixel)	Synthetic image	
							Error in V (pixel)	False vectors
Reference	60.66	449.4	6.2	29.2	13.5	1.0	0.07	0.0%
1	60.03	444.8	6.0	29.8	13.4	1.0	0.07	0.0%
2	60.29	443.4	2.1	29.6	6.2	8.2	0.39	6.7%
3	60.36	442.9	1.2	29.7	4.2	9.0	0.44	7.1%
4	61.23	447.3	2.7	33.6	−0.6	38	2.2	60%
5	61.23	446.8	2.1	32.1	−0.6	41	2.2	70%
6	60.89	446.6	5.7	29.7	−0.3	49	2.8	74%
7	61.57	451.3	4.5	30.5	−0.5	87	3.4	95%
8	60.32	435.4	−1.2	30.2	0.0	126	—	100%
9	60.88	448.9	21.7	46.2	−1.9	144	—	100%
10	60.28	456.3	25.9	32.4	7.8	166	—	100%
11	61.44	458.3	16.3	34.7	6.9	178	—	100%
12	62.68	465.2	12.6	8.6	0.1	182	—	100%
13	61.23	433.0	3.8	33.9	−8.2	263	—	100%
14	62.82	447.9	−13.6	30.6	−1.6	298	—	100%
15	63.51	444.6	3.0	32.3	−13.6	351	—	100%
16	62.11	482.1	7.6	26.0	−2.8	502	—	100%

Table 2 Errors after self-calibration

Calibration	Residual disparity (pixel)	Position error (pixel)	Synthetic image	
			Error in V (pixel)	Reconstruction error (pixel)
Reference	0.004	0	0.023	0.008
1	0.040	0.073	0.024	0.008
2	0.017	0.076	0.023	0.008
3	0.036	0.107	0.024	0.008
4	0.050	0.181	0.025	0.008
5	0.048	0.153	0.024	0.008
6	0.043	0.024	0.027	0.008
7	0.047	0.195	0.026	0.008
8	0.055	0.406	0.035	0.008
9	0.074	0.312	0.030	0.009
10	0.095	0.189	0.031	0.010
11	0.051	0.544	0.046	0.008
12	0.101	0.561	0.030	0.010
13	0.038	1.335	0.087	0.008
14	0.286	3.275	0.212	0.018
15	0.103	1.480	0.100	0.010
16	0.073	0.771	0.052	0.009

mapping function after equalizing the still different coordinate origins, in-plane rotations and conversion scales from millimetres to pixels. This positional deviation to the reference mapping function is shown in the third column of Table 2. For higher initial disparities, the final positional errors are also larger, but, in most cases, still acceptable. While the disparity signifies if the vectors for camera 1 and camera 2 are computed at the same position, which is most important for the errors of the final velocity field in regions of strong gradients, the positional errors relate to general residual warping of the coordinate system. The vectors are then computed at an incorrect x/y position (typically < 2 pixels) and with incorrect spatial derivatives (typically $< 1\text{--}3\%$). Given a typical distance between vectors of 8–16 pixels, a sub-pixel positional error in the coordinate system is still acceptable.

Finally, using the corrected mapping functions, the 2D3C stereo-PIV vector fields are computed and compared to the true displacement field. Table 2 shows the deviation from the true result, as well as the average reconstruction error. The reconstruction error originates partly from normal correlation errors, secondly due to vectors from cameras 1 and 2 computed at different positions and, lastly, due to slightly incorrect spatial derivatives of the mapping function, as might be the case for calibration 14.

For most cases, the reconstruction error is well below 0.02 pixels. The differences relative to the true velocity field are, therefore, mainly due to a residual disparity and warping of the in-plane x – y world coordinate system. The results for calibrations 13, 14 and 15 show the limit of the self-calibration procedure for large disparities ($1/3$ of the image size). Interestingly, calibration 16 is more accurate again. Looking at the projection angles, this might be due to the fact that this view is greatly deviated from, but still almost parallel to, the light sheet,

while calibrations 13–15 are more tilted and cover more depth in the observation volume.

For all other calibrations, the errors in V are less than 0.04 pixels, which is usually much smaller than what can be expected for real experiments with all other known error sources. The value of 0.023 pixels for the reference calibration is mostly due to the effect of image degradation due to warping and dewarping of the original synthetic particle image. At the same time, the results validate the accuracy of the basic stereo-PIV correlation and reconstruction algorithms. The multi-pass algorithm with simultaneously dewarped and deformed interrogation windows proves to be highly accurate.

The results show the potential of the self-calibration procedure to correct even large misalignments between the calibration plate and the light sheet. A small aperture has been used for a large depth of focus, and for the calibrations with the largest disparities, some parts of the images are already slightly out-of-focus. This suggests that the camera pinhole model is sufficiently accurate, as long as images are recorded within the depth of focus.

3.2 Experimental results with translated plate

A flat target of size 5×7 cm with a random dot pattern has been translated in the z direction using a translation stage with an accuracy of about $5 \mu\text{m}$. The plate has been moved in the z direction by 1–5 mm. The calibration has been done using a 3D calibration plate recorded at eight views at arbitrary positions, which have been evaluated by bundle adjustment.

The experiment has been performed in air as well as in water, where the $z=0$ mm plane was about 45-mm deep in water behind a 3-mm glass plate. In Table 3, the main camera pinhole parameters after self-calibration

Table 3 Mapping function parameter

	Initial disparity (pixel)	Residual disparity (pixel)	Camera	x_0/y_0 (pixel)	s_x	f (mm)	T_x (mm)	T_y (mm)	T_z (mm)	R_x (°)	R_y (°)	R_z (°)
Air, Eight views bundle	65	0.07	1	586/552	0.998	60.8	16.1	1.3	411.7	1.4	20.3	2.7
			2	706/548	0.998	60.9	9.2	0.2	435.2	-0.2	-24.3	2.2
Water, Eight views bundle	7.9	0.16	1	-1882/560	0.957	81.0	113.9	-10.3	537.6	2.7	27.0	4.4
			2	4074/491	0.924	83.4	-162.0	-8.3	570.5	-1.7	-34.1	4.2

onto the random pattern target are shown. In both air and water, the small average residual disparity after self-calibration is mainly due to uncertainties in the peak detection of the correlation peak. The angle between the cameras is about 45°.

For water, the calculated pinhole parameters are quite different. The principal point lies far outside the chip and the skew factor is much less than 1, as would be expected for square pixels. This is due to the distortions of the air–glass–water interface, which can only be fitted by the camera pinhole model by non-physical values for s_x , f and the principal point. Nevertheless, the calibration as a whole seems to be very accurate, which is indicated by the fact that, for the bundle adjustment in water, all calibration points in space were fitted with an accuracy better than 0.1 pixels.

In Table 4 the results of the measured displacements are shown. The second row shows, for air, the displacements using a triangulation and plane-fit method, as in the standard self-calibration procedure. First, the disparity of the images recorded at e.g. position $z=1$ mm are computed relative to the corrected mapping function at $z=0$, and, using triangulation, the position of points on the $z=1$ mm plane are determined, and a plane is fitted through those points. The measured position of the $z=1$ –5 mm planes agree with the translation stage movement to within 4 μ m, which proves a high accuracy of the mapping function throughout the volume. This agrees well with the measurement errors of 2–4 μ m reported by Fournel et al. (2004) for the same type of experiment.

The third row shows the stereo-PIV evaluation in air using the corrected 8-view bundle calibration. The vector computation was performed with multi-pass iterations and deformed interrogation windows of size 64×64 pixels. The measured displacements in air agree with the uncertainties of the translation stage for 1-mm

and 2-mm displacements. For larger displacements, there seems to be a systematic bias. Given the good agreement of the triangulation and plane fit method using the same mapping function, this error is probably due to the fact that the 3C reconstruction uses mapping function derivatives calculated only at the $z=0$ plane. These derivatives change over the quite large distance of 5 mm (equal to 100 pixels). A better way would be to compute the 3C reconstruction using the more accurate triangulation method. Further investigations are needed to look at this effect in detail. For practical purposes, given a bias of 0.06 mm = 1.2 pixels at $dz=5$ mm = 100 pixels would lead to a bias of 0.06 pixels for typical displacements of 5 pixels, which is commonly less than due to all other error sources.

The rms in the other u components and v components remain always around 0.03 pixels. While the average v displacement is zero everywhere, the u components show a bias of 0.36 pixels at $dz=1$ mm, increasing linearly with larger displacements. Closer inspection revealed that the axis of the translation stage was not exactly perpendicular to the random pattern plate, resulting in a slight x movement.

The experiment has also been repeated with a translation in the x direction. Again, the measured displacements agree well with the settings from the translation stage within the uncertainties of the translation stage.

In water, the results are similar. For up to 2 mm (40 pixels), the accuracy is high, with some bias effects for larger displacements. The rms values are larger than in air due to a residual incorrect warping of the coordinate system, which, e.g. for $dz=1$ mm, shows up as a gradient across the image in the displacement on the order of 0.03 pixels. The triangulation and plane fit recovers accurately the movement of the plate by the translation stage. This proves that the camera pinhole model and the self-calibration method remains suffi-

Table 4 Comparison of measured displacements with true displacements

Translation stage moved by	$dz=1$ mm	$dz=2$ mm	$dz=3$ mm	$dz=4$ mm	$dz=5$ mm
Air triangulation	0.999 mm	2.002 mm	2.996 mm	4.002 mm	5.001 mm
Air stereo-PIV	0.995 ± 0.001 mm	1.992 ± 0.003 mm	2.973 ± 0.003 mm	3.962 ± 0.004 mm	4.941 ± 0.006 mm
	20.22 ± 0.03 pixel	40.46 ± 0.06 pixel	60.38 ± 0.05 pixel	80.47 ± 0.08 pixel	100.35 ± 0.12 pixel
Water triangulation	1.001 mm	2.005 mm	2.995 mm	3.998 mm	4.992 mm
Water stereo-PIV	0.997 ± 0.004 mm	1.995 ± 0.006 mm	2.977 ± 0.007 mm	3.969 ± 0.007 mm	4.949 ± 0.009 mm
	20.59 ± 0.06 pixel	41.18 ± 0.13 pixel	61.45 ± 0.15 pixel	81.92 ± 0.15 pixel	102.15 ± 0.19 pixel

ciently accurate, even with strong distortions from refractive index changes.

3.3 Experimental results with particle images

An experiment has been performed in air with water droplets of a few microns in size. Two jets of air are passing upwards with high speed between two cylinders of 10-mm diameter, generating high shear regions. The field-of-view is about 85×70 mm at a distance of 500 mm. The camera CCD array is $1,280 \times 1,024$ pixels. The angle between the cameras is about 50° . Three calibrations (1–3) are taken using a 3D calibration plate. The average disparity relative to the recorded particle images is 44, 82 and 252 pixels, respectively. The disparity map for calibration 2 is shown in Fig. 5c, which corresponds to a rotational misalignment around the y axis of about 13° , together with a z displacement of around 5 mm.

Vector fields are computed for the high shear region (Fig. 5a). The final interrogation window size is 32×32 pixels with 75% overlap. The average velocity gradient is 3%, with maximum values around 20% close to the jets. Without self-calibration correction, large displacement errors are visible and many vectors are removed due to reconstruction errors larger than 1 pixel, as shown in Fig. 6a for calibration 1. For calibrations 2 and 3, no meaningful vector fields can be calculated.

Self-calibration has been performed using interrogation window sizes of 128×128 pixels with an overlap of 50% and a summation of 16 correlation maps, which provided correlation peak positions with an accuracy of better than 0.1 pixels, as can be deduced from the laser plane fit error, which is less than 0.1 pixels for all three mapping functions. After equalizing the coordinate origin, the direction of the x axis, and the global scale from pixels to millimetres between the three corrected mapping functions, the average difference in the mapped x/y position between the three corrected mappings is less than 0.4 pixels, as calculated directly from the functional form of the mapping functions.

With self-calibration correction, the vector fields of the three calibrations are almost identical (Fig. 6b–d with background colour = vorticity). The average vector difference between calibration 1 and 3 in the high shear region is only 0.055 pixels (Fig. 6e). Note that the displayed vectors are enlarged by a factor of 50.

4 Self-calibration into closed measurement volumes

In many applications, it is difficult or impossible to perform an accurate calibration inside the measurement volume. Here, it is necessary to calibrate from the outside and, somehow, compute a corrected mapping function for the measurement plane inside the volume using the disparity map and an appropriate

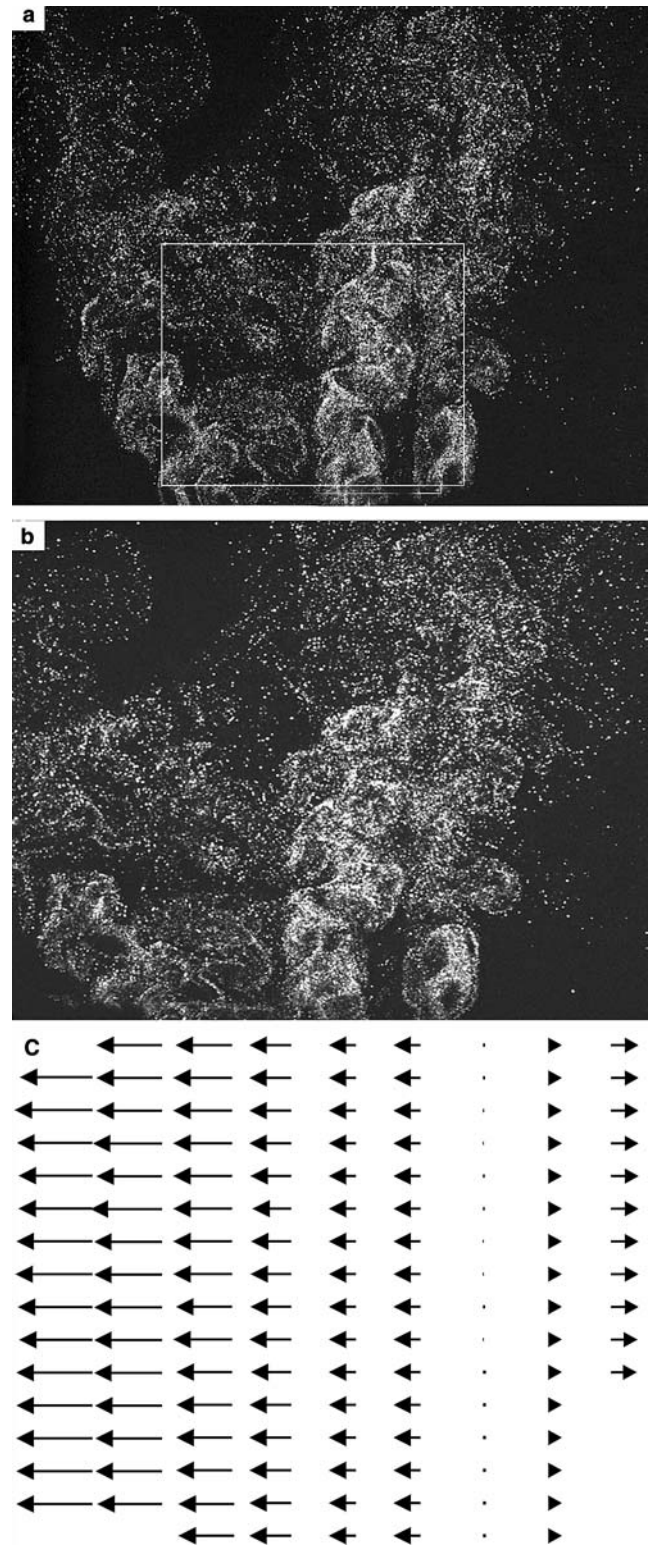


Fig. 5a–c Particle image viewed by camera 1 (a) and 2 (b) dewarped with calibration 2. The white rectangle defines the high shear area evaluated. c Corresponding disparity map typical of a rotation around the y axis

correction scheme. Different strategies are investigated in Sects. 4.1 and 4.2, together with experimental verification.

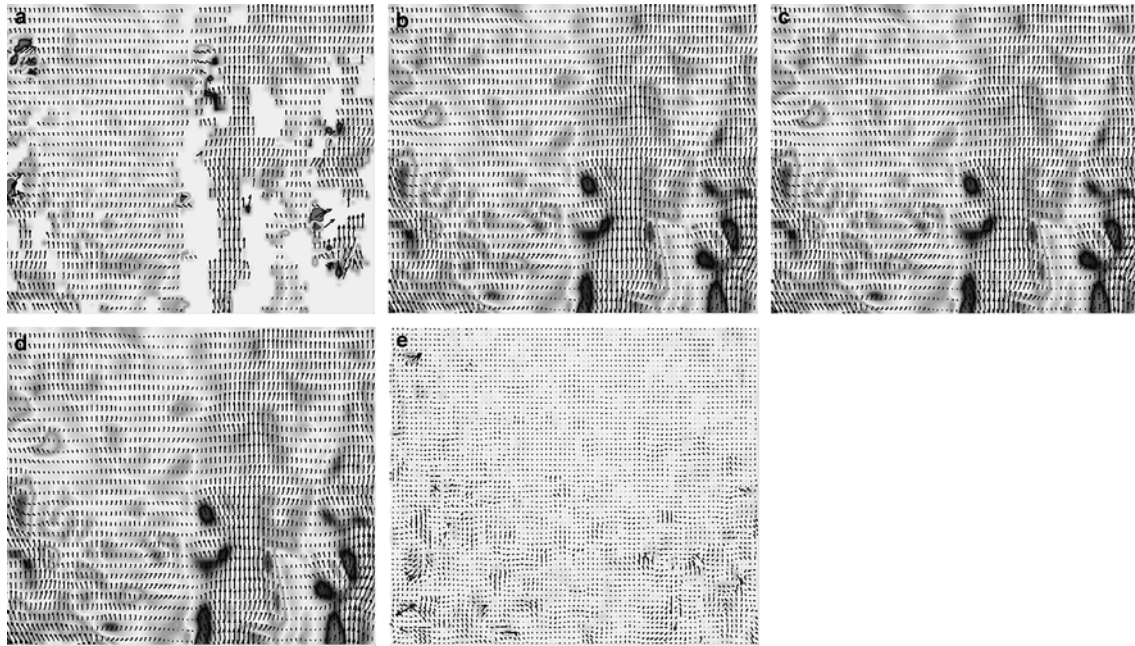


Fig. 6 **a** Vector field for uncorrected calibration 1. Background colour = vorticity. **b-d** Velocity fields after self-calibration for calibrations 1–3. **e** Difference between vector field of calibration 1 and 3. The vectors in **e** are enlarged by 50x. Field-of-view is about 600x500 pixel

4.1 Calibration outside with similar optical setup

Given that, as shown in Sect. 3.2, the camera pinhole model without modifications can handle a refractive index change with sufficient accuracy, a straightforward strategy is to perform the calibration outside under conditions as similar to the real measurement conditions as possible. As shown in Fig. 7 (left and middle), this can be done by first focussing the cameras onto the light sheet plane, then sufficiently retracting both cameras with a translation stage such that a small water basin can be placed in front of the water channel and performing a calibration inside the water basin in the standard way with a single or multiple views of a 2D calibration or 3D calibration plate. It is important that the distance between the cameras and the front side of the water basin is the same as it is relative to the front side of the water channel in the real measurement position (L in Fig. 7). Finally, the cameras are moved back to the original

position, the real experiment is performed and the standard self-calibration procedure can be applied to correct the mapping function onto the light sheet. With an accurate mechanical setup, the accuracy of this approach is the same as if both the calibration and the recording were done inside the measurement volume.

A scan in the z direction through the measurement volume can be done by moving the laser light sheet to a new z position and computing the self-calibration separately for each z position. If the travel distance is larger than the depth of focus, it is necessary to move the cameras and the light sheet simultaneously. In this case, it is required to perform a calibration outside in the water basin for each z scan position separately by adjusting the distance between the cameras and the front side of the water basin accordingly.

4.2 Calibration outside in air

Of course, it would be easier to perform the calibration outside in air without a water basin and to self-calibrate onto the recorded light sheet in water (Fig. 7 right). The results using this method are shown in Table 5. This approach leads to large errors for the standard pinhole

Fig. 7 *Left*: recording position. *Middle*: calibration procedure outside but in a similar optical setup. *Right*: calibration outside in air and self-calibration onto recording using 3-media model

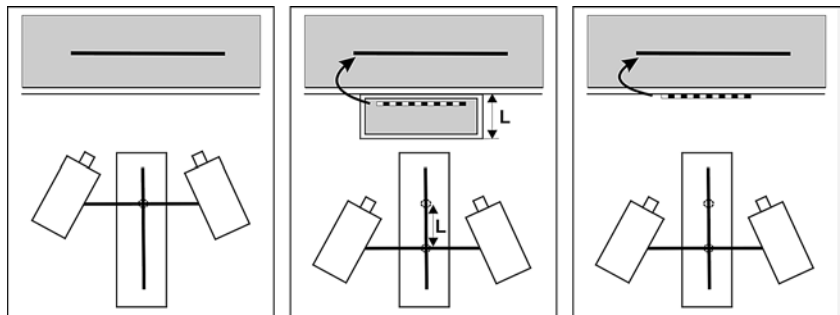


Table 5 Measured displacements for calibration in air and self-calibration to recording in water

Translation stage moved by		$dz = 1$ mm	$dz = 2$ mm	$dz = 3$ mm	$dz = 4$ mm	$dz = 5$ mm
Standard model	Triangulation	0.737	1.491	2.228	2.978	3.720
	Stereo-PIV	0.743 ± 0.004	1.486 ± 0.003	2.217 ± 0.006	2.957 ± 0.007	3.688 ± 0.006
3-media model	Triangulation	0.996	2.005	3.003	4.014	5.016
	$dZ_{g-1} = 40$					
	Stereo-PIV	1.006 ± 0.004	2.012 ± 0.004	3.001 ± 0.006	4.001 ± 0.006	4.989 ± 0.015
	$dZ_{g-1} = 40$					
	Stereo-PIV	1.006 ± 0.003	2.009 ± 0.004	2.996 ± 0.005	3.994 ± 0.006	4.981 ± 0.007
	$dZ_{g-1} = 30$					
	Stereo-PIV	1.003 ± 0.003	2.007 ± 0.004	2.994 ± 0.005	3.991 ± 0.006	4.976 ± 0.007
	$dZ_{g-1} = 20$					
	Stereo-PIV	0.972 ± 0.003	1.944 ± 0.005	2.899 ± 0.006	3.865 ± 0.007	4.819 ± 0.009
	$dZ_{g-1} = 0$					
	Stereo-PIV	1.021 ± 0.007	2.041 ± 0.013	3.045 ± 0.018	4.059 ± 0.025	5.061 ± 0.032
	$dZ_{g-1} = 40, \alpha_x = 10^\circ$					
	Stereo-PIV	1.000 ± 0.001	2.001 ± 0.003	2.995 ± 0.005	3.978 ± 0.006	4.960 ± 0.008
	$dZ_{g-1} = 40, \alpha_y = 10^\circ$					

Displacements given in millimetres, 1 mm = 21.1 pixels

Distance between glass plate and light sheet dZ_{g-1} is 40 mm

model. The z displacement is shortened by about the index-of-refraction of water. During self-calibration, it is not possible to refit all parameters of the pinhole model to the distorted water case. Especially, the internal camera parameters, like s_x and f , are not changed during the standard self-calibration procedure. In principle, the disparity map, as defined by the 3×3 fundamental matrix equation, has 8 degrees of freedom and only the 3-plane parameter need to be refitted by the standard self-calibration method. Hence, one has the extra degrees of freedom to fit parameters like s_x or the relative camera orientation. But there are too many parameters to be fitted, so one must be restricted to a subset given by the particular experimental setup.

A better approach is to modify the camera pinhole model to accommodate the air–glass–water interface. This method allows an accurate physically motivated model. A 3-media model (e.g. air–glass–water) has been implemented according to Maas (1996). Using Snell’s law and an iterative approach, the bending of light rays through glass and water is calculated. The thickness and refractive index of each medium can be specified. The thickness dZ_{g-1} of the last medium (water) is defined by the distance between the light sheet plane and the previous medium (glass). Currently, this value must still be measured, together with the two angles between the light sheet and the glass plate. The distance between light sheet and glass plate can be measured, for example, by focussing on the light sheet with a large aperture (small depth of focus) and then traversing the cameras backwards until a target mounted on the front side of the water channel is in focus.

The initial calibration in air is done without the 3-media model, which is switched on for self-calibration (triangulation step) and the subsequent stereo-PIV vector computation.

The results are shown in Table 5. With the 3-media model, the movement of the random pattern plate is

accurately computed both by the triangulation method and the stereo-PIV evaluation. For larger displacements, again, some deviations are visible, showing the limits for the 3-media model, both for the triangulation and the stereo-PIV method. This is not relevant for typical PIV experiments, since $dz = 5$ mm corresponds to an extremely large displacement of 100 pixels. The only drawback is that one has to know the distance dZ_{g-1} between the light sheet and the glass plate. In the current experiment, dZ_{g-1} is 40 mm, which was measured conventionally. Ideally, one could also fit this from the disparity map, but initial tests indicate that the fit algorithm then becomes unstable and is not able to fit the particle plane and the glass plate at the same time. Further work is needed to explore this possibility.

In Table 5, the results of the self-calibration procedure with assumed wrong distances dZ_{g-1} to determine the sensitivity of inaccurate measurements of the position of the laser sheet, with respect to the glass plate, are also shown. The computed stereo-PIV displacements remain accurate to within 1% for offsets of dZ_{g-1} up to 20 mm. The important feature of the 3-media model is that the z derivatives of the mapping functions, which are off by a factor given by the index-of-refraction of water, are again accurately calculated. Any remaining in-plane x disparities are compensated by the self-calibration procedure. Larger errors are visible for wrongly measured tilts of the light sheet. For an assumed tilt around the y axis of 10° , dz and dy ($=0$) are accurate to within 2%, but there is a systematic error in dx of 3%. For a tilt around the x axis there is a systematic error in dy of 8% and residual y disparities of a few pixels, which can not be compensated for by self-calibration. Of course, the 3-media model is also of advantage when calibrating in-situ in water. It has been verified that the camera pinhole parameters return to physically meaningful values in comparison to the strange ones as shown in Table 3.

5 Laser light sheet thickness and relative position

As an added benefit of the correction scheme, the thickness and relative position of the two laser sheets of the double-pulse PIV laser can be deduced from the correlation maps. The correlation peaks are smeared out due to particles contributing throughout the light sheet, as shown in Fig. 8. Consequently, the light sheet thickness can be computed by simple geometric considerations from the correlation peak width. When the cross-correlation is done on dewarped images, the correlation peak width is given by:

$$w_c = d \left[\frac{1}{\tan(\alpha_1)} + \frac{1}{\tan(\alpha_2)} \right] \quad (8)$$

with d being the light sheet thickness and α_1 and α_2 are the viewing angles of cameras 1 and 2 relative to the x axis in the case when the cameras are placed horizontally along the x axis. This is assuming point particles. For real particles, the correlation peak is folded with the particle point spread function, which could be calculated by auto-correlation. The width of the correlation peak in units of pixels is a function of the ratio of the thickness of the light sheet in relation to the distance between the camera and the light sheet. For a typical stereo-PIV experiment with measured x - y displacements of 5–10 pixels, one needs a light sheet thickness at least twice as thick to measure z components of the same order. Therefore, typical correlation peak widths are on the order of 10–20 pixels.

If this analysis is done for both laser light sheets separately, the position of the two planes can be compared to determine the overlap of the two light sheets and the flatness of each sheet. Another method often used for determining the overlap of the two laser sheets—besides visual inspection—is by setting the dt between the two laser shots as short as possible. Then,

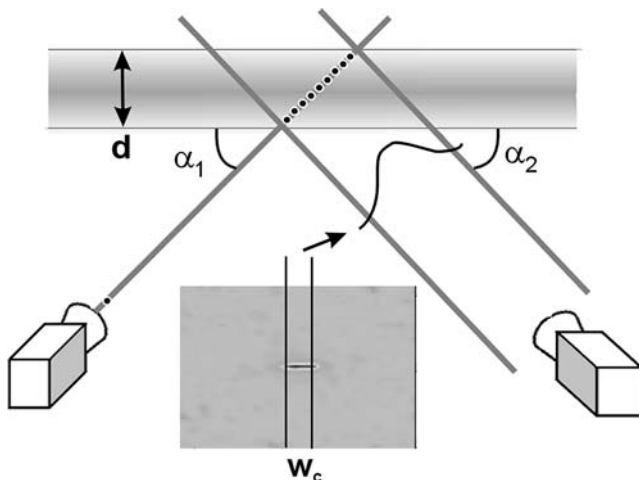


Fig. 8 Particles throughout the light sheet contribute to the correlation peak. From the peak width, the light sheet thickness can be computed

the two images of each camera show almost the same particle pattern (if the light sheets are well aligned) and a cross-correlation gives a high correlation coefficient, which is indicative of good light sheet overlap. The above method based on the cross-correlation of cameras 1 and 2 is computationally more intensive, but offers the advantage of showing the real position in space of each laser beam, together with a clear indication of which screw in the laser head should be adjusted.

6 Summary

A self-calibration correction scheme has been developed to compensate for misalignment between the calibration plate and the light sheet. After fitting a camera pinhole model to a 3D calibration plate, a disparity vector map is calculated by cross-correlating the dewarped particle images of cameras 1 and 2 taken at the same time. For higher stability and accuracy, the correlation maps of many image pairs are summed up. The disparity vectors are used to calculate world points on the real measurement plane by triangulation. A plane is fitted through these points.

Finally, the mapping functions are transformed to the new plane. It is shown that this calibration scheme provides highly accurate mapping functions with final displacement errors smaller than is expected from the other error sources, like the basic particle image velocimetry (PIV) correlation algorithm for real images. This has been confirmed for different experimental setups. It has been shown that, with such a correction, the $z=0$ plane of the mapping function lies within 0.1 pixels of the middle of the light sheet.

The self-calibration scheme is advisable in any case to check the calibration and to improve the accuracy. Since it works well even for very large misalignments, it eliminates the need for an alignment of the calibration plate with the light sheet, which is often difficult and time consuming.

A modified 3-media camera pinhole model has been implemented to account for index-of-refraction changes along the optical path. It is then possible to calibrate from the outside, e.g. a closed water channel, and self-calibrate onto the recordings inside the channel. This method allows stereo-PIV measurements inside closed measurement volumes, which was not previously possible.

As a side benefit, the correlation maps can be analysed to yield the position and thickness of the two laser sheets and, therefore, the degree of overlap and flatness of each sheet.

References

- Calluaud D, David L (2004) Stereoscopic particle image velocimetry measurements of the flow around a surface-mounted block. *Exp Fluids* 36:53–61

- Coudert S, Schon JP (2001) Back-projection algorithm with misalignment corrections for 2D3C stereoscopic PIV. *Meas Sci Technol* 12:1371–1381
- Fournel T, Coudert S, Lavest JM, Collange F, Schon JP (2003) Self-calibration of telecentric lenses: application to bubbly flow using moving stereoscopic camera. In: *Proceedings of the 4th Pacific symposium on flow visualization and image processing (PSFVIP-4)*, Chamonix, France, June 2003
- Fournel T, Lavest JM, Coudert S, Collange F (2004) Self-calibration of PIV video cameras in Scheimpflug condition. In: Stanislas M, Westerweel J, Kompenhans J (eds) *Particle image velocimetry: recent improvements. Proceedings of the EURO-PIV 2 workshop*, Zaragoza, Spain, March/April 2003. Springer, Berlin Heidelberg New York, pp 391–405
- Hartley R, Sturm P (1994) Triangulation. In: *Proceedings of the ARPA image understanding workshop*, Monterey, California, November 1994, pp 957–966
- Hartley R, Zissermann A (2000) *Multiple view geometry in computer vision*, 1st edn. Cambridge University Press, Cambridge, UK
- Maas HG (1996) Contributions of digital photogrammetry to 3D PTV. In: Dracos T (ed) *Three-dimensional velocity and vorticity measuring and image analysis techniques*. Kluwer, Dordrecht, The Netherlands, pp 191–208
- Meinhart CD, Wereley ST, Santiago JG (1999) A PIV algorithm for estimating time-averaged velocity fields. In: *Proceedings of optical methods and image processing in fluid flow*, 3rd ASME/JSME joint fluids engineering conference, San Francisco, California, July 1999
- Prasad AK (2000) Stereoscopic particle image velocimetry. *Exp Fluids* 29:103–116
- Soloff SM, Adrian RJ, Liu ZC (1997) Distortion compensation for generalized stereoscopic particle image velocimetry. *Meas Sci Technol* 8:1441–1454
- Tsai RY (1986) An efficient and accurate camera calibration technique for 3D machine vision. In: *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR'86)*, Miami, Florida, June 1996, pp 364–374
- Willert C (1997) Stereoscopic digital particle image velocimetry for application in wind tunnel flows. *Meas Sci Technol* 8:1465–1479